Data Science

CSE558

Recap

- Markov's Inequality

$$\mathbf{P}\left(\mathbf{X} \geq \mathbf{a}\right) \leq \frac{\mathbf{E}[\mathbf{X}]}{\mathbf{a}}$$

- Chebyshev's Inequality

$$\mathbf{P}\left(|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq \mathbf{a}\right) \leq \frac{\mathbf{Var}[\mathbf{X}]}{\mathbf{a}^2}$$

Overview

- Concentration Inequalities

Example

Suppose your friend gave a job interview. The company received 4,000 applicants for the interview, from which they will take the best 300 people. They conducted an aptitude test to judge their abilities. The mean score and standard deviation of the test are 56 and 6, respectively. If your friend had scored 80, then do you think she will be getting the job?

Moment Generating Function (m.g.f.)

The m.g.f. of a random variable $\mathbf{X}$ is, $M_{\mathbf{X}}(\mathbf{t}) = \mathbf{E}[e^{\mathbf{tX}}]$

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbf{E}[e^{\mathbf{tX}}] = \sum_{i=0}^{\infty} \frac{\mathbf{t}^i \cdot \mathbf{E}[\mathbf{X}^i]}{i!} = 1 + \mathbf{t} \cdot \mathbf{E}[\mathbf{X}] + \frac{\mathbf{t}^2 \cdot \mathbf{E}[\mathbf{X}^2]}{2!} + \dots$$

$$\frac{dM_{\mathbf{X}}(\mathbf{t})}{d\mathbf{t}} = \frac{d}{d\mathbf{t}}\left[1 + \mathbf{t} \cdot \mathbf{E}[\mathbf{X}] + \frac{\mathbf{t}^2 \cdot \mathbf{E}[\mathbf{X}^2]}{2!} + \dots\right]$$

$$= \left[0 + \mathbf{E}[\mathbf{X}] + \frac{2\mathbf{t} \cdot \mathbf{E}[\mathbf{X}^2]}{2!} + \dots\right]$$

$$= \left[0 + \mathbf{E}[\mathbf{X}] + \frac{2\mathbf{t} \cdot \mathbf{E}[\mathbf{X}^2]}{2!} + \dots\right]\Bigg|_{\mathbf{t}=0} = \mathbf{E}[\mathbf{X}]$$

Moment Generating Function (m.g.f.)

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$$\left.\frac{dM_{\mathbf{X}}(\mathbf{t})}{d\mathbf{t}^2}\right|_{t=0} = \left.\frac{d}{d\mathbf{t}^2}\left[1 + \mathbf{t} \cdot \mathbf{E}[\mathbf{X}] + \frac{\mathbf{t}^2 \cdot \mathbf{E}[\mathbf{X}^2]}{2!} + \dots\right]\right|_{t=0} = \mathbf{E}[\mathbf{X}^2]$$

$$\left.\frac{dM_{\mathbf{X}}(\mathbf{t})}{d\mathbf{t}^k}\right|_{t=0} = \mathbf{E}[\mathbf{X}^k]$$

For two independent random variables $\mathbf{X}$ & $\mathbf{Y}$,

$$M_{\mathbf{X}+\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{X}}(\mathbf{t}) \cdot M_{\mathbf{Y}}(\mathbf{t})$$

Chernoff Bound

For a random variable $\mathbf{X}$ and $\mathbf{a} \in \mathbf{R}$,

$$\mathbf{P}(\mathbf{X} > \mathbf{a}) \leq \min_{\mathbf{t}>0} \frac{\mathbf{E}[e^{\mathbf{tX}}]}{e^{\mathbf{at}}} \quad \text{and} \quad \mathbf{P}(\mathbf{X} < \mathbf{a}) \leq \min_{\mathbf{t}<0} \frac{\mathbf{E}[e^{\mathbf{tX}}]}{e^{\mathbf{at}}}$$

Very powerful!

Gives exponentially decreasing bounds.

Example

A biased coin that lands heads with probability 0.2 is flipped 100 times independently. With what probability it land heads at most 50 times

Solution: Consider the random variable, $\mathbf{X} = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$

Now, $\mathbf{E}\left[e^{\sum_{i=1}^{100} \mathbf{t}\cdot\mathbf{X}}\right] = \mathbf{E}\left[\prod_{i=1}^{100} e^{\mathbf{t}\cdot\mathbf{X}}\right] = \prod_{i=1}^{100} \mathbf{E}\left[e^{\mathbf{t}\cdot\mathbf{X}}\right] = \prod_{i=1}^{100} \left[0.2 \cdot e^{t\cdot 1} + (1-0.2)e^{t\cdot 0}\right]$

$$= \prod_{i=1}^{100} \left[1 + 0.2(e^{t\cdot 1} - 1)\right] \leq \prod_{i=1}^{100} \left[e^{(e^{\mathbf{t}}-1)\cdot 0.2}\right] = e^{(e^{\mathbf{t}}-1)\cdot 20}$$

$$\mathbf{P}\left(\#\mathbf{heads} > 50\right) \leq \frac{\mathbf{E}[e^{\sum_{i=1}^{100} \mathbf{t}\cdot\mathbf{X}}]}{e^{50\cdot\mathbf{t}}} \leq \frac{e^{(e^{\mathbf{t}}-1)\cdot 20}}{e^{50\cdot\mathbf{t}}} = \left(\frac{e^{(e^{\mathbf{t}}-1)}}{e^{2.5\cdot\mathbf{t}}}\right)^{20}$$

Example

A biased coin that lands heads with probability 0.2 is flipped 100 times independently. With what probability it land heads at most 50 times

Solution: Consider the random variable, $\mathbf{X} = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$

$$\mathbf{P}\left(\#\mathbf{heads} > 50\right) \leq \left(\frac{e^{(e^{\mathbf{t}}-1)}}{e^{2.5\cdot\mathbf{t}}}\right)^{20}$$

Setting **t** = ln(2.5) we get,

$$\mathbf{P}\left(\#\mathbf{heads} > 50\right) \leq \left(\frac{e^{(e^{\mathbf{t}}-1)}}{e^{2.5\cdot\mathbf{t}}}\right)^{20} = \left(\frac{e^{2.5-1}}{2.5^{2.5}}\right)^{20} = \frac{1.35}{10^{7}}$$

Chernoff Bound on Sum of Poisson Trials

Let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ be independent poisson trials, such that $\Pr(X_i = 1) = p_i$

Let, $\mu = \mathbf{E}[X] = \mathbf{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbf{E}[X_i] = \sum_{i=1}^{n} p_i$

$$\mathbf{Pr}(|X - \mu| \geq \delta \cdot \mu)$$

$$M_{X_i}(t) = \mathbf{E}[e^{tX_i}] = p_i e^t + (1 - p_i) = 1 + p_i(e^t - 1) \leq e^{p_i(e^t - 1)}$$

$$M_X(t) = \prod_{i=1}^{n} M_{X_i}(t) \leq \prod_{i=1}^{n} e^{p_i(e^t - 1)} = \exp\left\{\sum_{i=1}^{n} p_i(e^t - 1)\right\} = e^{(e^t - 1)\mu}.$$

Apply Markov's inequality for any t > 0, $\Pr(X \geq (1 + \delta)\mu) = \Pr(e^{tX} \geq e^{t(1+\delta)\mu})$

Chernoff Bound on Sum of Poisson Trials

For any *t > 0*

$$\Pr(X \geq (1+\delta)\mu) = \Pr(e^{tX} \geq e^{t(1+\delta)\mu}) \leq \frac{\mathbf{E}[e^{tX}]}{e^{t(1+\delta)\mu}} \leq \frac{e^{(e^t-1)\mu}}{e^{t(1+\delta)\mu}}.$$

For *δ > 0*, setting *t = ln(1+δ) > 0* we get,

$$\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}}\right)^{\mu}$$

For any *t < 0*

$$\Pr(X \leq (1-\delta)\mu) = \Pr(e^{tX} \geq e^{t(1-\delta)\mu}) \leq \frac{\mathbf{E}[e^{tX}]}{e^{t(1-\delta)\mu}} \leq \frac{e^{(e^t-1)\mu}}{e^{t(1-\delta)\mu}}$$

For 0 < *δ < 1*, setting *t = ln(1-δ) < 0* we get,

$$\Pr(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}}\right)^{\mu}$$

Special Cases

Let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ be independent random variables such that for every 1 ≤ i ≤ n,

$\Pr(X_i = 1) = \Pr(X_i = -1) = \frac{1}{2}$ let, $X = \sum_{i=1}^{n} X_i$

For any *a > 0,* prove that, $\Pr(|X| \geq a) \leq 2\mathrm{e}^{-a^2/2n}$

Proof: $\mathbf{E}[\mathrm{e}^{tX_i}] = \frac{1}{2}\mathrm{e}^{t} + \frac{1}{2}\mathrm{e}^{-t} = \sum_{i \geq 0} \frac{t^{2i}}{(2i)!} \leq \sum_{i \geq 0} \frac{(t^2/2)^i}{i!} = \mathrm{e}^{t^2/2}.$

$$\mathbf{E}[\mathrm{e}^{tX}] = \prod_{i=1}^{n} \mathbf{E}[\mathrm{e}^{tX_i}] \leq \mathrm{e}^{t^2 n/2}$$

Now, applying Markov’s inequality,

$$\Pr(X \geq a) = \Pr(\mathrm{e}^{tX} \geq \mathrm{e}^{ta}) \leq \frac{\mathbf{E}[\mathrm{e}^{tX}]}{\mathrm{e}^{ta}} \leq \mathrm{e}^{t^2 n/2 - ta}$$

Special Cases

Let $X_1, X_2, \ldots, X_n$ be independent random variables such that for every $1 \leq i \leq n$,

$\Pr(X_i = 1) = \Pr(X_i = -1) = \frac{1}{2}$ let, $X = \sum_{i=1}^{n} X_i$

For any $a > 0$, prove that, $\Pr(|X| \geq a) \leq 2e^{-a^2/2n}$

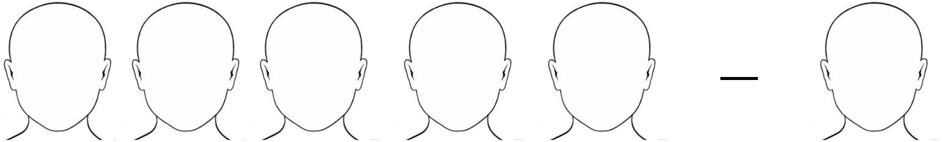
Proof: Applying Markov's inequality,

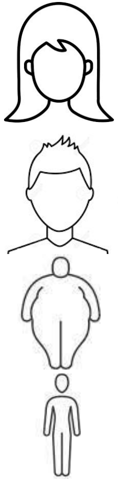
$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{\mathbf{E}[e^{tX}]}{e^{ta}} \leq e^{t^2 n/2 - ta}$$

Setting, t = a/n we get,

$$\Pr(X \geq a) \leq e^{-a^2/2n} \quad \text{and} \quad \Pr(X \leq -a) \leq e^{-a^2/2n}$$

Set Balancing





1	0	1	0	1	—	1
0	1	0	1	0	—	0
0	0	1	1	0	—	1
1	1	0	0	1	—	0

Dataset = A

#subjects = m

#features = n

Partition: $b \in \{-1,+1\}^m$

$$\min_{b\in\{-1,+1\}^m} \max_{i\in[n]} |b^T a_i|$$

Or, equivalently

$$\min_{b\in\{-1,+1\}^m} \|Ab\|_\infty$$

Problem: Partition the subject into treatment group (T) and control group (C) such that each feature has a roughly equal number of subjects in each group.

Set Balancing

$$\min_{b \in \{-1,+1\}^m} ||Ab||_\infty$$ Brute force takes $O(2^m)$ time.

Approximate solution?

For every $1 \leq i \leq m$, b_i is -1 or +1 with equal probability.
Takes $O(m)$ time.

Compute, $\Pr(||Ab||_\infty > k)$

Set Balancing - Analyze Solution

$$\min_{b \in \{-1,+1\}^m} \|Ab\|_\infty$$

Approximate solution: For 1 ≤ i ≤ n, b_i is -1 or +1 with equal probability.

Let a_i be the i^{th} row of A having k 1's.

If $k \leq \sqrt{4m \ln n}$ then $|b^T a_i| \leq \sqrt{4m \ln n}$

If $k > \sqrt{4m \ln n}$ then consider the random variable $Z_i = b^T a_i$

Now applying Chernoff's inequality to bound Z_i,

$$\Pr\left(|Z_i| > \sqrt{4m \ln n}\right) \leq 2e^{-4m \ln n / 2k} \leq \frac{2}{n^2}$$

Now taking a union bound, $\mathbf{Pr}\left(\|Ab\|_\infty \geq \sqrt{4m \ln(n)}\right) \leq \frac{2}{n}$